

Preventive Maintenance — Electric Motor

Healthy Bearing — Routine Inspection and Lubrication
Assets: AST_MTR_001 (Conveyor Motor A), AST_MTR_002 (Conveyor Motor B)
Bearing Types: SKF6310 | Est. Duration: 2 h | Crew: 1–2 | Priority: LOW
SOP_002 | v1.1 | Revised: 2026-03-01

1. Purpose and Scope

This procedure covers routine preventive maintenance for electric motor bearings that are confirmed healthy by the Monitoring Agent (anomaly score below 0.30, all signals within $\pm 2\sigma$ of baseline). Actions are limited to visual inspection, grease top-up, and vibration and temperature confirmation. No mechanical disassembly or bearing replacement is performed under this SOP. If the inspection reveals unexpected findings, STOP and escalate to SOP_001. Applies to SKF6310 bearings on AST_MTR_001 and AST_MTR_002.

2. Prerequisites

- Monitoring Agent anomaly score ≤ 0.30 — confirmed in latest monitoring record
- All vib, kurtosis, and temp signals within $\pm 2\sigma$ of baseline_registry values
- Planned maintenance slot available — LOW priority, can be deferred to next planned stop
- PART_004 (Synthetic Grease) available in BIN_L1 for top-up if required
- LOTO_001 current (AST_MTR_001) or LOTO_005 current (AST_MTR_002) — required if accessing grease nipple
- One technician with LOTO_CERTIFIED credential
- Portable vibration meter and infrared thermometer

3. Safety Requirements

3.1 LOTO Isolation Procedure

LOTO Required	Yes — for grease nipple access (motor cover panel must be opened)
Permit — AST_MTR_001	LOTO_001 Isolation: MCC-12, Valve-4
Permit — AST_MTR_002	LOTO_005 Isolation: MCC-13, Valve-5
Energy Type	Electrical
Verification	Attempt start after lock applied. Confirm motor does not start.

3.2 PPE and Zone Requirements

Mandatory PPE	Electrical insulation gloves, safety helmet, steel-toed boots
Zone — AST_MTR_001	C — Controlled approach with standard PPE
Zone — AST_MTR_002	A — Normal approach; standard PPE applies

4. Tools and Materials Required

Part Number	Description	Quantity	Bin Location
PART_004	Synthetic Grease — for top-up only if due	As required (approx. 30–40 g)	BIN_L1

- Grease gun loaded with PART_004
- Infrared thermometer
- Portable vibration meter (mm/s RMS, handheld)
- Torque screwdriver (for panel fasteners if applicable)

- Clean lint-free cloth for external surface inspection

5. Step-by-Step Procedure

1	Confirm anomaly score ≤ 0.30 from latest monitoring record before proceeding.
2	Apply LOTO per Section 3.1. Verify zero energy.
3	Inspect motor exterior: check for oil or grease leaks, loose fasteners, unusual surface temperatures, or physical damage. Record findings.
4	Inspect grease nipple and purge point for correct seating and no signs of contamination entry.
5	Dispense grease slowly. Observe purge point — when fresh grease appears at purge, stop dispensing. Do not exceed 40 g total.
6	Inspect purged grease on lint-free cloth: light amber, clean, smooth texture = acceptable. Dark, gritty, metallic particles, or burnt smell = STOP and escalate to SOP_001.
7	Remove LOTO. Controlled restart.
8	Take vibration reading at 30 minutes: record vib_rms_mm_s and temp_c.
9	Compare readings against baseline (BRG_001/BRG_002 baseline mean: ~ 2.0 mm/s, $\sim 57^{\circ}\text{C}$; BRG_003/BRG_004: ~ 1.8 mm/s, $\sim 52^{\circ}\text{C}$).
10	If readings within $\pm 1\sigma$ of baseline: QA PASS. Record on work order and close.
11	If readings above $\pm 2\sigma$: escalate to SOP_001 for full inspection.
12	Update bearing_master.last_lubrication_date and confirm current_health_score in system.

6. Critical Specifications

Lubricant Type	PART_004 — Synthetic Grease (Shell Gadus S2 V220 or equivalent)
Top-up Quantity	Approximately 30–40 g maximum. Stop when fresh grease appears at purge.
Re-lube Interval	Every 60 days under normal operating conditions
Grease Condition Rejection Criterion	Dark colour, metallic particles, acrid smell, or gritty texture → reject, escalate to SOP_001

7. Post-Maintenance Acceptance Criteria

Parameter	AST_MTR_001 (BRG_001/002)	AST_MTR_002 (BRG_003/004)	Failure Action
Vibration RMS (at 30 min)	≤ 2.5 mm/s	≤ 2.2 mm/s	Escalate to SOP_001
Bearing Temperature	$\leq 62^{\circ}\text{C}$	$\leq 55^{\circ}\text{C}$	Check grease fill; re-inspect
Kurtosis (historian, 30 min)	≤ 2.8	≤ 2.5	Escalate to SOP_001
Signal Quality Score	≥ 0.97	≥ 0.97	Check CH_001/002 or CH_003/004

8. Common Issues and Troubleshooting

- If purged grease is contaminated → immediate escalation to SOP_001; bearing must be inspected for internal damage.
- If vibration rises after re-greasing → excess grease; allow 30 min of operation for grease to distribute before re-measuring.
- If grease does not appear at purge after 40 g dispensed → purge path blocked; do not over-pump. Open for inspection.
- If kurtosis above 2.5 at 30 min → not a lubrication-only issue; escalate for full fault assessment.

9. Related Documents

- SOP_001 — Outer Race Bearing Replacement (escalation target)
- SOP_006 — Post-Repair QA Procedure
- CASE_005 — Healthy check, AST_MTR_002, BRG_003, no fault found
- CASE_007 — Preventive check, AST_MTR_001, BRG_002 (NDE), no fault found
- CASE_008 — Preventive check, AST_MTR_002, BRG_004 (NDE), no fault found